Hereditary Diffuse Gastric Cancer

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Genetic syndrome first reported among Maori population in New Zealand

- Autosomal dominant
- Early-onset diffuse gastric cancer ($\tilde{x}$ age = 40)
- High penetrance

CDH1 mutations and HDGC

Pathologic variants in CDH1 gene associated with HDGC

Penetrance in Populations Enriched in HDGC

Prophylactic Gastrectomy

- Stanford

90% of patients undergoing prophylactic gastrectomy harbor microscopic foci of gastric cancer in their stomach

Endoscopic Screening - NIH

A large NIH cohort study of 270 CDH1 carriers undergoing surveillance with systematic gastric mapping found only two patients developed clinically evident pT3N0 gastric carcinomas over a median 31-month follow-up, with no gastric cancer deaths.

(Asif et al. 2023)

Endoscopic Screening - Cambridge

imilarly, a 16-year longitudinal cohort study demonstrated that endoscopic surveillance with systematic random biopsies can identify early lesions and inform decision-making about gastrectomy.

(Lee et al. 2023)

Endoscopic Screening

2025 study of 20 carriers undergoing long-term surveillance (mean 62 months) found no patients developed aggressive (greater than pT1a) cancer, even among those with positive pT1a biopsies who continued surveillance.

aside: Expert Endoscopic Surveillance in CDH1 Pathogenic Variant Carriers Seems Safe, Even After Positive (pT1a) Biopsies. Journal of Medical Genetics. 2025. Benusiglio PR, Leenhardt R, Duros C, et al.New

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